

MODULE 2

Equipment Mastery

**Recognise the truck. Match the load.
Ask the right questions before you book.**

Dry vans, box trucks, reefers, flatbeds, step decks, power only and Sprinter vans — with real photos, spec sheets with dimensions, the weight and size rules that decide what is legal, and the questions that stop you booking the wrong load.

8

equipment spec
sheets

11

outdated claims
corrected

~7 hours

study + practice

50+

checks with answers

How to use this module

In Module 1 you learned who is who. Now you learn **what** moves the freight. Every load you will ever consider starts with one check: **does this freight physically and operationally fit this truck?**

Study rhythm (about 7 hours)

- **Session 1 (1.5 h):** the basics — anatomy, weight, size. Read slowly; everything else builds on it.
- **Session 2 (2 h):** dry van and box truck. Learn the spec sheets until you can draw them.
- **Session 3 (2 h):** reefer, flatbed, step deck, power only, Sprinter.
- **Session 4 (1.5 h):** matching, codes, mistakes, the practice postings, the visual quiz and the check.

How to read a spec sheet

Each equipment type has a one-page spec sheet: a **real photo**, a **side drawing with dimensions**, a **spec table**, the **questions to ask before booking**, and its current **DAT equipment types**. Numbers are **typical**. Trucks differ, so the dispatcher's habit is always: *confirm with the carrier*.



Corrections to your old manuals

Your equipment PDFs are visually excellent but contain several outdated or over-simplified claims (fixed payloads, "13'6" federal height", "4 chains per load", "pre-cool 2 hours", old Sprinter codes, CDL by box length). Each correction is flagged with a red box in this module.

Contents

L2.0 The basics: anatomy, weight and size rules	4
L2.1 Dry van — 53' and 48'	8
L2.2 Box truck / straight truck	13
L2.3 Reefer	17
L2.4 Flatbed and step deck	20
L2.5 Power only, Sprinter vans and special cases	25
Matching, codes, mistakes and the equipment checklist	28
Practice, visual quiz and module check	32
Answer key · Glossary · Sources	35

After this module you can

- Recognise 8 equipment types from a photo
- Explain why "80,000 lb" is not the cargo weight
- Work out pallet counts and freight height limits
- Pick the right equipment for a load posting
- Ask the 4–5 questions each equipment type needs
- Avoid the four classic equipment booking mistakes



2

MODULE 2 · EQUIPMENT & FREIGHT MATCHING

Wrong equipment = wrong load

A 53' van at a 48'-only dock. A reefer set to the wrong temperature. A box truck with no liftgate at a house. A flatbed without the tarps the load needs. Each mistake costs a load, a fee or a claim — and each is prevented by knowing the equipment and asking one question.

No rates in this module. The old manuals printed "\$ per mile" for each truck type. Rates change with market, lane and season — they are taught with live data in Module 7.

Eight kinds of equipment you will dispatch

Eight kinds of equipment you will dispatch

All real photos. Learn to recognise each one on sight — and know what it can and cannot carry.



Dry van

Enclosed general freight



Box truck

One chassis, local, liftgate



Reefer

Temperature-controlled



Flatbed

Open deck, secured + tarped



Step deck

Lower deck for taller freight



Power only

Tractor, no trailer



Sprinter van

Small, fast, urgent



Curtainside

Van that opens at the side

Three big families

- **Enclosed:** dry van, reefer, box truck, Sprinter van, curtainside. Freight is protected from weather; loaded from the rear (or side, for curtainside).
- **Open deck:** flatbed, step deck and specialised trailers. Freight that is too big, heavy or awkward for a box; loaded by forklift or crane; must be secured and often tarped.
- **Capacity only:** power only — a tractor that pulls someone else's trailer.



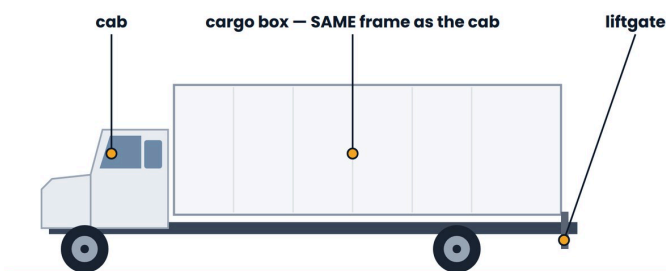
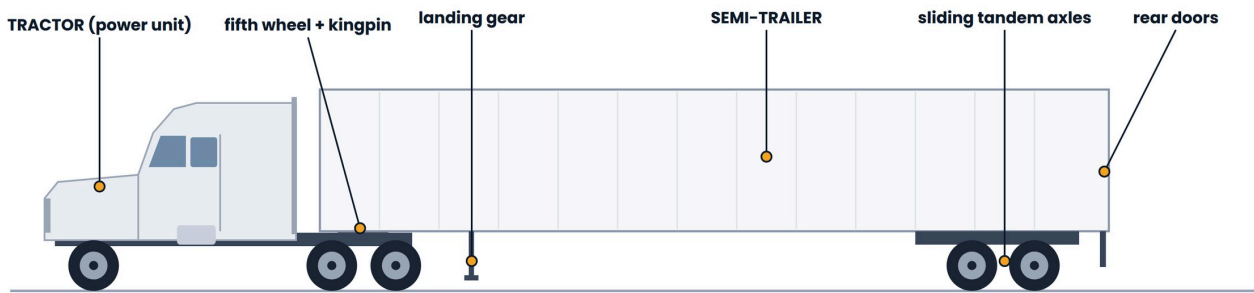
The first check on every load

Before rate, before broker, before anything: **can this truck carry this freight — size, weight, temperature, loading method and gear?** If the answer is “no” or “not sure,” you are not ready to call.

Which equipment will you dispatch most? In Pakistani dispatch offices, beginners usually start with **dry van, box truck and power only**. Reefer and flatbed often come later because they add temperature and securement risk.

Tractor-trailer vs straight truck

TRACTOR-TRAILER two units — the tractor can drop one trailer and hook another



STRAIGHT TRUCK

one vehicle, one chassis
cannot drop its box

Words you must use correctly

Term	Meaning
Tractor (power unit)	The truck with the engine and cab. Pulls a trailer.
Semi-trailer	The trailer. Its front rests on the tractor's fifth wheel.
Fifth wheel / kingpin	The coupling. The trailer's kingpin locks into the tractor's fifth wheel.
Landing gear	Legs that hold the trailer up when it is dropped.
Tandem axles	Two axles together. Trailer tandems often slide to balance weight.
Straight truck	Cab and cargo box on one frame, e.g. a box truck.

Why the difference matters to you

A tractor can **drop** one trailer and **hook** another. That makes **drop-and-hook** loads and **power-only** work possible. A straight truck cannot drop its box: it must wait while it is loaded and unloaded.

Weight and licensing also differ. Tractor-trailers are combination vehicles (Class A CDL). Box trucks are single vehicles whose licence depends on their weight rating (page 14).



Three weight words

GVWR = the maximum a vehicle is rated to weigh, loaded (on a door sticker). **Tare** = what it weighs empty.

Payload = what it can carry = the legal maximum minus the tare.

80,000 lb is the whole truck, not the cargo

The weight truth: 80,000 lb is for the **WHOLE** truck

Federal Interstate limits: 80,000 lb gross · 20,000 lb single axle · 34,000 lb tandem axle · bridge formula for axle spacing



Example: $80,000 - 19,000 - 15,000 = 46,000$ lb maximum cargo, on paper

Tractor + fuel + driver $\approx 19,000$

Empty trailer $\approx 15,000$

Room for cargo $\approx 46,000$ (theoretical)

Why real loads are usually lighter (often 42,000–45,000 lb for a dry van):

each axle group has its own limit, the freight may not sit evenly, trucks and trailers weigh different amounts, and some roads or bridges have lower limits. **Rule: ask the carrier how much THIS truck can legally take.**

The federal numbers

On the Interstate system, FHWA's rules are: **80,000 lb gross** (unless the bridge formula gives less), **20,000 lb on a single axle**, **34,000 lb on a tandem axle**. The **bridge formula** limits weight on groups of axles according to how far apart they are, to protect bridges.

So the cargo is whatever remains after the **tractor, fuel, driver and trailer** are counted, and only if it is spread so that no axle group is overweight.



Correction: "Dry van max payload = 45,000 lb" / "Flatbed max = 48,000 lb"

These are rough real-world examples, **not legal limits**. A heavy tractor and a steel trailer carry less; a light tractor and an aluminium trailer carry more. Your old manual also claimed "mountain areas lower payload to 30–40k" — there is no such general rule.



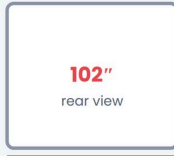
The dispatcher habit

Every truck gets a **"max weight"** in your notes, confirmed by the carrier: *"Truck 12 legal up to 44,500."* Never book to a generic number from a PDF.

Width, length and height: who sets the limit?

8'6"

Width: 102 inches



Federal maximum on the National Network. Wider freight = over-dimension, which needs permits.

Federal rule

48' / 53'

Trailer length

48' min • 53' common

States must allow semitrailers of at least 48' on the National Network. 53' is the most common van length and widely allowed.

Federal minimum + state rules

No federal limit

Height

14' in some states
13'6" in many states



States set height limits — most between 13'6" and 14'. Low bridges and routes matter. Never teach "13'6" is the federal limit".

State / route rule

What FHWA says

- **Width:** 102 inches (8'6") is the federal maximum on the National Network. Wider = over-dimension.
- **Length:** states must allow semitrailers of **at least 48'** on the National Network; many allow 53' (and some longer, grandfathered).
- **Height:** "There is no Federal vehicle height requirement." States set their own — most between **13'6" and 14'**.



Correction: "13'6" is the legal height"

There is **no single federal height limit**. 13'6" is common in the East; 14' is common in many Western states; bridges and tunnels can be lower. Flatbed and step-deck freight must be checked against the route.



What this means for vans

Vans are built to about 13'6" tall so they can run almost everywhere. Height becomes your problem mainly with **open-deck** freight (pages 22–24).

The 53-foot dry van



53-foot dry van

The workhorse of U.S. freight: an enclosed trailer for general, non-temperature freight.

Outside	53' long · 8'6" (102") wide · about 13'6" high
Inside	≈ 52'6" L · 99–101" W · 108–110" H
Rear door	Swing doors ≈ 98" wide × 108–110" high
Pallets (48"×40")	26 loaded straight · up to 30 turned
Volume	≈ 3,800–4,000 cubic feet
Trailer weight	≈ 13,000–15,000 lb empty
Cargo weight	Often ≈ 42,000–45,000 lb — ask the carrier
Loads from	The rear, at a dock (floor ≈ 4' high)

Before booking, the dispatcher asks

- Weight and pallet count — do they fit this truck?
- Floor-loaded or palletized? Stackable?
- Any length limit ("48' max") at the shipper/receiver?
- Air-ride, team, straps or load bars required?

DAT equipment types: **V** **VA** **VM** **VC** **VG** **VZ**

Typical figures — they vary by manufacturer and year. * No federal height limit; states set it (usually 13'6"–14'). Confirm the actual truck with the carrier.



Read the sheet like a dispatcher

The two numbers that decide most van loads are **pallet positions (26)** and **legal weight**. A load of 26 light pallets fills the floor long before it reaches the weight limit: it **cubes out**. A load of 18 pallets of paper can reach the weight limit with floor space left: it **weighs out**.

Vans load at **docks**. The trailer floor sits about 4 feet off the ground, the same height as a warehouse dock, so forklifts drive straight in.

What dry vans haul

Palletised or boxed goods that need weather protection but **no temperature control**: retail and grocery dry goods, paper, packaging, beverages, electronics, household goods, auto parts — like the printed boxes on load #12354.

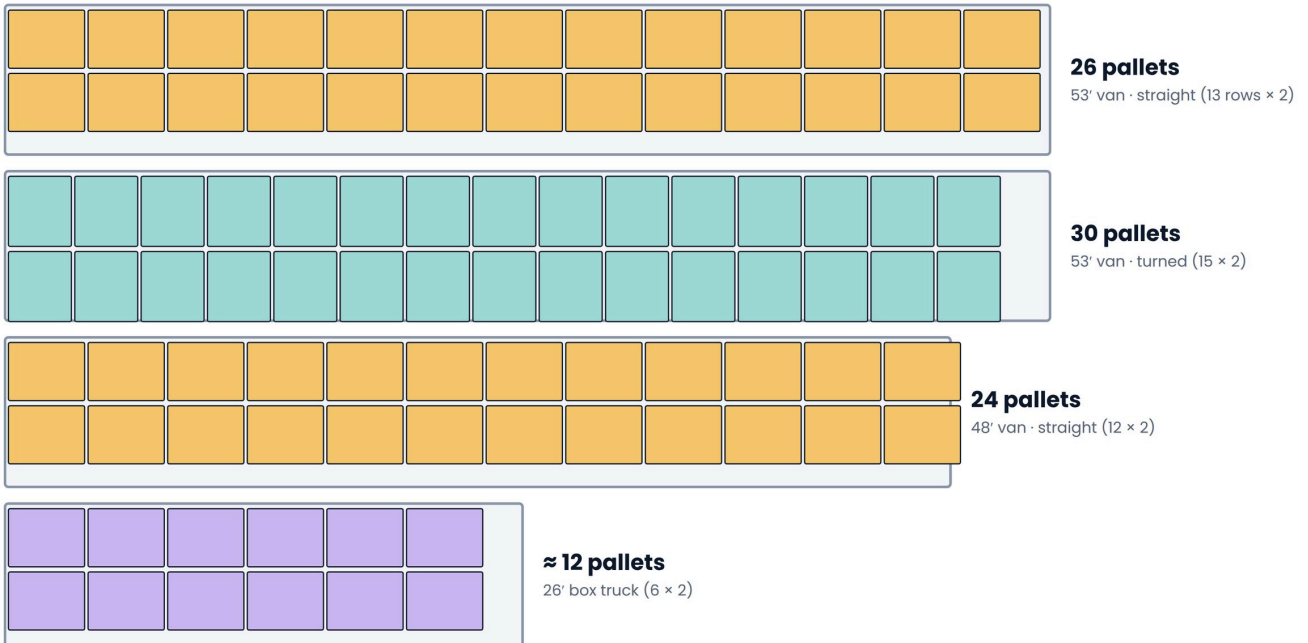
What they can't

Frozen or temperature-sensitive food (reefer), freight taller than about 9' or wider than the doors, and freight that must be loaded from the top or side with a crane (flatbed / curtainside).

Pallet math: cube vs weight

Pallet math: how many 48" × 40" pallets fit?

Top view, rear doors on the right. Standard U.S. pallet = 48" × 40". Weight often runs out before floor space does.



Straight vs turned

A standard U.S. pallet is **48" × 40"**. Loaded **straight** (48" side along the trailer), two pallets fit side by side and 13 rows fit in a 53' van: **26 pallets**. **Turned** (40" along the trailer, 48" + 48" across), 15 rows fit: **up to 30**, but only if the trailer is wide enough inside and the shipper loads it that way.



Worked example

A posting says: **"24 pallets, 42,000 lb, 53' dry van."** Floor: $24 \leq 26$ ✓. Weight: 42,000 lb — you must ask the carrier whether this truck is legal at 42,000. If the truck's max is 43,500 ✓. If it is 41,000 ✗.

Questions that uncover problems

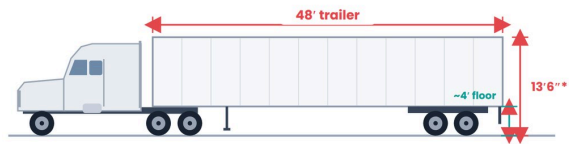
- **"Palletised or floor-loaded?"** Floor-loaded (loose boxes) takes hours to load and unload by hand.
- **"Stackable?"** Some pallets can be double-stacked, which changes the space needed.
- **"Pallet size?"** Not every pallet is 48 × 40; oversized pallets mean fewer fit.
- **"Exact weight?"** Remember load #12354: the RC said 2,000 lb, the BOL said 5,729 lb.



Say it back

"So that's **24 pallets, non-stackable, 42,000 pounds**, dock to dock." Reading back catches mistakes on the call itself.

The 48-foot dry van — and when length matters



48-foot dry van

Shorter van for tight docks, city streets and some older facilities.

Outside	48' long · 8'6" (102") wide · about 13'6" high
Inside	≈ 47'6" L · 98–100" W · 108–110" H
Pallets (48"×40")	24 straight · up to 28 turned
Volume	≈ 3,300–3,500 cubic feet
Cargo weight	Similar to a 53' — weight, not length, is usually the limit
When it matters	Posting says "48' max", tight urban dock, short yard
Federal rule	States must allow semitrailers of at least 48' on the National Network

Before booking, the dispatcher asks

- Does the load need 53' of space or only weight capacity?
- Is there a length limit at pickup or delivery?
- Is the 48' van also air-ride / team if required?

DAT equipment types: V VA

Typical figures — they vary by manufacturer and year. * No federal height limit; states set it (usually 13'6"–14'). Confirm the actual truck with the carrier.

Why 48' still exists

Older docks, tight city streets, some yards and some customers were built around shorter trailers. A few postings say **"48' max"** or **"no 53'"**. Most carriers today run 53' vans; some also have 48'.



Classic mistake: wrong length

The receiver only accepts 48' trailers. Your 53' shows up and is turned away at the gate: no delivery, redelivery fees, a furious broker. **Ask:** "Any length restriction at pickup or delivery?"

Question	48' van	53' van
Pallets (straight / turned)	24 / up to 28	26 / up to 30
Weight capacity	Similar — depends on the truck	Similar — depends on the truck
Fits "48' max" locations	✓	✗



Quick check

A load is **25 pallets, 20,000 lb, "48' max at delivery."** Can your 48' van take it? No: 25 pallets needs more than 24 straight positions, unless the shipper can turn them. Ask the broker how the pallets are loaded.

Inside the van: gear, doors and variants

Inside a van: securement gear you must recognise

Real photos. FMCSA cargo-securement rules apply to enclosed vans too — freight must not shift, tip or fall.



Load bars

Bars wedged wall-to-wall to stop freight sliding



E-track / logistic posts

Rails in the walls for straps and bars



Straps

Hold freight to E-track in vans and box trucks



Airbags (dunnage bags)

Fill gaps between pallets



Blocking & bracing

Wood built around cargo so it cannot move

Dispatcher question: “Does the load need straps, load bars or blocking — and does this truck actually carry them?”

Securement applies to vans too

FMCSA’s cargo-securement rules apply to **all cargo-carrying commercial vehicles** in interstate commerce, including enclosed vans. Freight that shifts can break through doors, tip the trailer or be damaged. Loads may require **straps, load bars or blocking**, and brokers sometimes ask for “2 load bars” or “straps required.”

Swing doors vs roll-up door

Most 53’ vans have **swing doors**. Some vans and many box trucks have a **roll-up door**, which sits inside the opening and can reduce usable door height. It rarely matters, until the freight is tall.



Curtainside van (DAT: VC)

Side curtains slide open so a forklift can load from the side. Used for rolls, long items, or where rear loading is impossible.



Correction: “Curtain-side is still filtered under V”

DAT now lists **Van w/ Curtains (VC)** as its own type. Use current platform categories, not old cheat sheets.

Van variants and what they really mean

DAT type	Name	What it means	When it matters
V	Van	Standard enclosed dry van	The default for general freight
VA	Van Air-Ride	Air-bag suspension gives a softer ride	When the posting / broker requires air-ride
VM	Van w/ Team	Two drivers take turns so the truck keeps moving	Time-critical loads that need team service
VC	Van w/ Curtains	Curtainside, side-loading	Side loading required
VG	Van Lift Gate	Van with a liftgate	No-dock delivery with a van
VZ	Van Hazmat	Van hauling placarded hazardous materials	Only if carrier + driver are hazmat-qualified
VP	Van w/ Pallet Exchange	Driver swaps empty pallets for loaded ones	Pallet exchange loads — ask how it works
VV	Van Vented	Van with vents for airflow	Some produce / commodities needing ventilation



Correction: “VA is mandatory for fragile freight”

Air-ride is a **load requirement when the posting, broker or shipper asks for it**, not an automatic rule for glass or electronics. Read the requirements and ask.



Correction: “VM = loads over 700 miles”

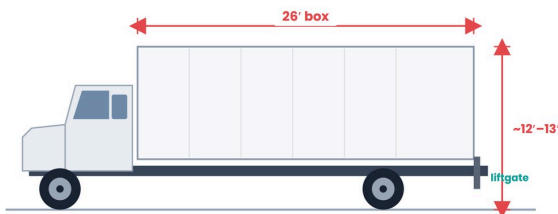
VM means a **van with a team**. No 700-mile rule exists. Use it when the load needs team service **and** your carrier has team drivers.



“VR — van or reefer”?

Older manuals teach VR. It did not appear in DAT’s current equipment list when this module was checked (Sep 2026), so it is not taught as a core code. Check the live platform in Module 6.

The straight box truck



26-foot straight box truck

Cab and cargo box on one chassis. Local and regional freight, residential and no-dock deliveries.

Box inside	≈ 26' L · about 8' W · about 8' H (varies a lot)
Pallets (48"×40")	About 12
GVWR	Often 25,999 lb (no CDL) — some are 26,001+ (CDL needed)
Cargo weight	Often ≈ 8,000–12,000 lb, depending on GVWR and truck weight
Liftgate	Common but not standard — check it exists and its capacity
Pallet jack	Needed to move pallets onto the liftgate at no-dock stops
Smaller sizes	16' ≈ 6 pallets · 20' ≈ 8 · 24' ≈ 10

Before booking, the dispatcher asks

- Dock or no dock? Residential? Liftgate + pallet jack needed?
- Weight and pallet count vs this truck's real payload
- Inside height and door type (roll-up doors reduce height)
- Does the driver need a CDL for this truck?

DAT equipment types: **SB** **SB-Reefer** **SB-Hazmat**

Typical figures — they vary by manufacturer and year. * No federal height limit; states set it (usually 13'6"–14'). Confirm the actual truck with the carrier.

Where box trucks shine

Shorter local and regional runs, partial loads, deliveries to **stores, homes, job sites and businesses without docks**, and freight too small for a 53' van. Many new small carriers and owner-operators start with a box truck, so you may dispatch them early in your career.

The numbers that trip beginners

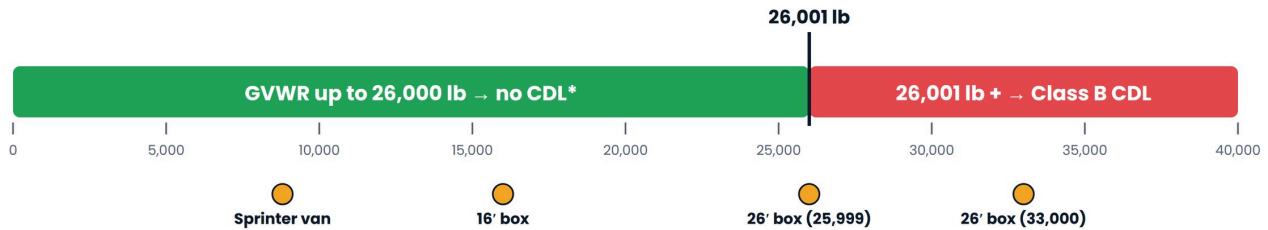
- **Payload is small** — often 8,000–12,000 lb, not 45,000.
- **Interior sizes vary** — ask for inside length, width, height and door opening.
- **Liftgate is not standard** — confirm it exists and its capacity.

Remember load #12354: a 26' box truck, and the BOL weight was nearly three times the RC weight. On a box truck that difference really matters.

Who can drive it? CDL and the 26,001 lb line

CDL depends on weight rating and use — not box length

Single vehicle (box truck, van): look at the GVWR on the door-jamb sticker.



Class A

Combination 26,001 lb+ where the towed unit is over 10,000 lb — e.g. every tractor-trailer.

* Class C exceptions

Smaller vehicles still need a CDL if they carry placarded hazmat or 16+ passengers.

Example GVWRs are typical, not guaranteed. Source: FMCSA CDL classes.

FMCSA's classes, in plain words

- **Class A:** combinations of 26,001 lb or more where the towed unit is over 10,000 lb — every tractor-trailer.
- **Class B:** a **single vehicle** with a GVWR of 26,001 lb or more — heavier box trucks.
- **Class C:** smaller vehicles that carry **placarded hazmat** or **16+ passengers**.

A box truck rated at **25,999 lb** and driven without hazmat usually needs **no CDL**. The same-size box rated at **33,000 lb** needs a **Class B**.



Correction: "14–22 ft = non-CDL, 24–28 ft = CDL"

The licence depends on the **GVWR and the use** (hazmat, passengers), **not the length of the box**. Two 26' trucks can need different licences.



Where to find the GVWR

The manufacturer's certification label on the **driver's door jamb**. Ask the carrier for it when you onboard a box truck and write it in your truck notes.

Dock or no dock? Liftgate and pallet jack



Dock stop

Warehouse dock built at trailer-floor height. Forklifts drive in and out of the trailer. Dry vans and reefers expect docks.

Any van or box truck can work



No-dock / residential stop

Ground level — a home, a store front, a job site. Freight comes down on a liftgate and is moved with a pallet jack.

Needs liftgate + pallet jack (or forklift on site)

The questions for every box-truck stop

- "Is there a **dock** at pickup and at delivery?"
- "Is it **residential** or a business with a dock?"
- "Is a **liftgate** required? Does the driver need a **pallet jack**?"
- "Is there a **forklift** on site, or does the driver unload?"
- "How heavy is the heaviest pallet?" (It must be within liftgate capacity.)



Reefer box truck (DAT: Straight Box Truck – Reefer)

Box trucks can also be refrigerated. Treat them like a reefer: setpoint, mode, pre-cool if required.



Correction: "VG is the box-truck liftgate code"

DAT lists VG as **Van Lift Gate**. Box trucks have their own types: **Straight Box Truck**, **Straight Box Truck – Reefer**, **Straight Box Truck Hazmat**. Treat the liftgate as a **capability you confirm**, not a code to memorise.

Box truck vs 53' dry van



Box truck



53' dry van trailer

Feature	Box truck (26' typical)	53' dry van
Units	One vehicle: cab + box on one frame	Tractor + separate trailer
Floor	About 12 pallets	26 pallets (up to 30 turned)
Cargo weight	Often 8,000–12,000 lb	Often 42,000–45,000 lb
Licence	No CDL if GVWR ≤ 26,000 (and no hazmat); Class B above	Class A CDL
Stops	Docks, homes, stores, job sites (with liftgate)	Mostly docks
Typical runs	Local / regional, partials, LTL-style	Regional / long haul, full truckload
DAT search	Straight Box Truck types + broad "Partial" van search	V (and variants)



DAT's own advice for small equipment

Run **two searches**: one with the specific equipment type (Straight Box Truck, Sprinter Van), and a broad **Van Standard / Van Specialized** search with **Partial** selected, because "brokers don't always post them under specific truck types."

The refrigerated trailer (reefer)



53-foot refrigerated trailer (reefer)

Insulated trailer with a diesel refrigeration unit on the front. Keeps freight frozen, chilled or at a set temperature.

Outside	53' long · 8'6" wide · about 13'6" high
Inside	≈ 52' L · 97–100" W · 100–104" H (insulation takes space)
Pallets (48"×40")	Usually 26
Temperature	Units typically cover about -20°F to +70°F
Trailer weight	Heavier than a van (insulation + unit + fuel)
Cargo weight	Often ≈ 40,000–43,000 lb — ask the carrier
Modes	Continuous (always running) or cycle / start-stop

Before booking, the dispatcher asks

- Setpoint temperature and mode (continuous or cycle)?
- Pre-cool required? Any temperature record needed?
- Frozen, chilled or produce? Load temperature?
- Unit fuel and service OK? Washout/clean trailer required?

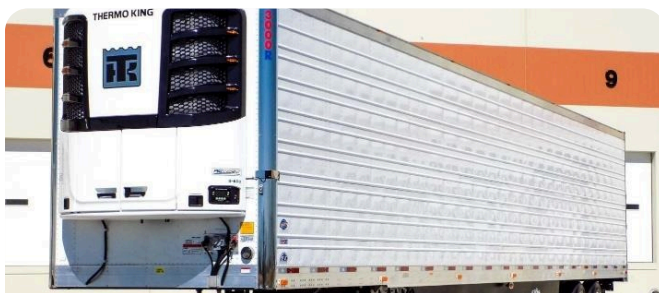
DAT equipment types: **R** **RA** **RM** **RZ** **RP**

Typical figures — they vary by manufacturer and year. * No federal height limit; states set it (usually 13'6"–14'). Confirm the actual truck with the carrier.



Why a reefer holds a bit less

Thick insulated walls, floor and roof make the inside **narrower and lower** than a dry van, and the unit, insulation and fuel make the trailer **heavier**, so it can carry less weight. Its floor has **grooves** so cold air can return to the unit underneath the pallets.



The unit on the front wall: a diesel engine and compressor with its own fuel tank.

Continuous vs cycle (start-stop)

Continuous: the unit runs all the time and air always moves, the usual choice for fresh produce and many sensitive loads. **Cycle / start-stop:** the unit switches off when the temperature is reached and restarts when needed. It saves fuel and is often used for frozen goods. **The shipper decides;** the RC or BOL usually says which.

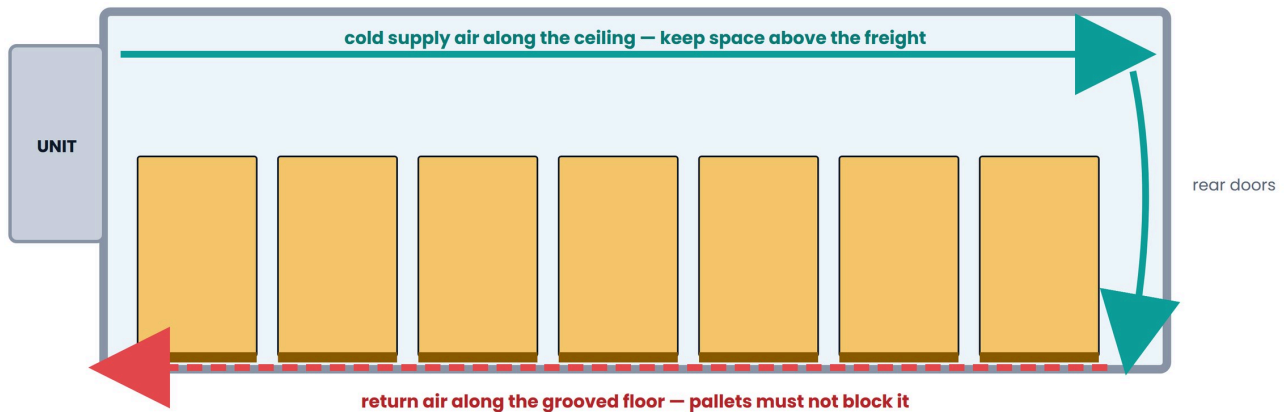


Inside: the air chute along the ceiling, insulated walls, grooved floor.

Cold air, and what “temperature control” means

How a reefer moves cold air

Side cutaway. Cold air leaves the unit at the top, travels to the rear, and returns along the floor. Blocked air = warm spots.



Dispatcher's job: confirm setpoint, mode and loading instructions — the shipper and driver handle the loading itself.

Three kinds of reefer freight (examples)

Type	Typical example	Watch for
Frozen	Ice cream, frozen meat — often around 0°F or below	Setpoint, cycle vs continuous, pre-cool
Chilled / fresh	Dairy, fresh meat — often mid-30s °F	Tight temperature windows
Produce	Fruit, vegetables — varies by product	Usually continuous; some products must not freeze

These are examples to build intuition. **The only temperature that counts is the one on the load instructions / RC.**



A reefer can often haul dry freight

With the unit off, a reefer is an insulated van. Carriers often take dry loads as backhauls. Ask the broker whether a reefer is acceptable for a van load, and check the extra trailer weight.



Correction: “No temperature log = no claim defence”

Too absolute. Temperature records are **important evidence**, and FDA's rule says carriers must be able to show they kept the requested temperature **when asked**. Outcomes of claims depend on the contract and the facts.

Before you book a reefer: five questions

Reefer workflow: follow the instruction, not a rule of thumb

FDA's sanitary-transport rule: the compartment must be adequately prepared, "including precooling, if necessary."



Instruction

Setpoint, mode, pre-cool, records — from the RC / shipper



Prepare

Clean trailer, fuel, pre-cool IF required



Verify

Unit set to the right °F and mode before loading



Load

Shipper loads; driver checks temp + count



Monitor

Check the unit en route; report alarms fast



Deliver

Temperature record ready if requested

✗ Old manual rule

"Always pre-cool 2 hours before loading."

✓ Correct rule

Pre-cool when the shipper / load instructions require it. Always verify the setpoint.

1. **"What is the setpoint?"** Write the exact number and °F. Never guess.
2. **"Continuous or cycle?"**
3. **"Is pre-cooling required, and to what?"**
4. **"Frozen, chilled or produce? What temperature is the product loaded at?"**
5. **"Any temperature recording or download needed at delivery?"**



Say it out loud — reefer read-back

"So that's **34 degrees Fahrenheit, continuous**, pre-cool to 34 before loading, fresh chicken, temperature download at delivery. Correct?"

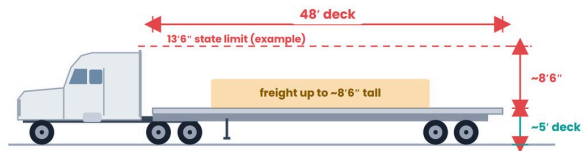


Correction: "Pre-cool the trailer 2 hours before loading"

No universal 2-hour rule exists. **Pre-cool when the shipper or the load instructions require it**, to the temperature they give, and always verify the setpoint before loading.

Why it matters: a wrong setpoint can mean a **rejected load** worth tens of thousands of dollars and a claim against your carrier.

The flatbed



48-foot flatbed

Open deck, no walls or roof. Loaded from the sides or top by forklift or crane. Freight must be secured and often tarped.

Deck	48' (also 53') long · 8'6" (102") wide
Deck height	About 5' from the ground
Freight height	About 8'6" if the state limit is 13'6"
Cargo weight	Often ≈ 45,000–48,000 lb (aluminum decks are lighter)
Typical freight	Steel, lumber, pipe, building materials, machinery
Gear	Straps, chains, binders, tarps, edge protectors, dunnage
Wider than 8'6"?	Over-dimension — permits and specialized planning

Before booking, the dispatcher asks

- Commodity, weight, dimensions (L × W × H)?
- Tarps needed? What size (4', 6', 8' drop)?
- Chains, straps, coil racks, edge protection?
- Crane or forklift? Side- or rear-load?

DAT equipment types: F FT FC FD FO

Typical figures — they vary by manufacturer and year. * No federal height limit; states set it (usually 13'6"–14'). Confirm the actual truck with the carrier.

Why flatbed is a different world

With a van, the box protects the freight. With a flatbed, **the driver builds the protection**: straps, chains, edge protectors, blocking and tarps. Loading is usually from the **sides or top** by forklift or crane, so the freight can be long, heavy or awkward.

Flatbed loads therefore depend on **dimensions, weight and gear**, not just the letter "F."

Tarps in one minute

- **Lumber tarps** — large, cover tall stacks.
- **Steel tarps** — smaller, for lower loads like coils or beams.
- **"Drop"** — how far the tarp hangs down the sides: 4', 6', 8'. A posting may say "8' drops required."
- **Smoke tarp** — covers the front of a load against exhaust.

DAT now shows handling requirements (tarps, chains, coil racks) on flatbed loads. Match them to what the truck carries.

Flatbed freight and gear

Flatbed freight: what it looks like on the road

Real photos. Flatbed loads are judged by dimensions, weight AND the securement gear the truck carries.



Lumber / building products

Commodity-specific rules apply



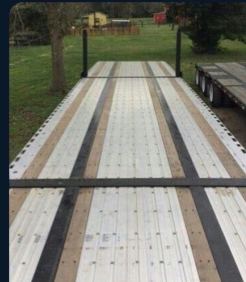
Pipe

Chocks, chains or straps per rules



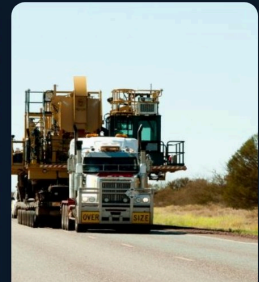
Tarped machinery

Tarps protect from weather



Empty deck + winches

Straps anchor to side winches



Over-dimension

Permits, escorts, route plans

Ask every time: commodity · weight · length × width × height · tarps (size) · chains / straps · edge protection · how it is loaded (forklift, crane, side, rear).

Gear checklist to confirm with the carrier

Gear	Used for
Straps + winches	Most general freight
Chains + binders	Steel, machinery, heavy items
Tarps (size, drop)	Weather protection when required
Edge protectors / corner guards	Protect straps and freight edges
Dunnage / lumber	Blocking, lifting freight off the deck
Coil racks	Steel coils (commodity-specific rules)



The division of responsibility

The driver and carrier are legally responsible for securing the load correctly. **The dispatcher** is responsible for not booking a load the truck can't secure, so confirm gear **before** saying yes.



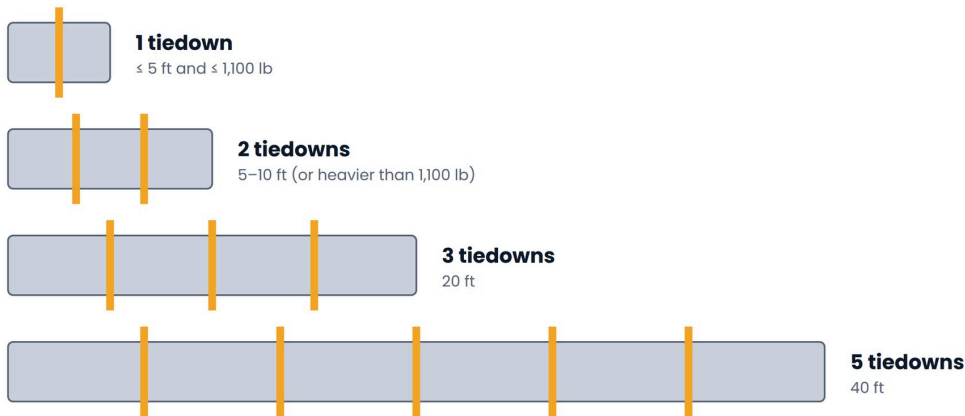
Over-dimension is a specialist area

Freight wider than 8'6", taller than the route allows, or heavier than legal limits needs **state permits**, sometimes escorts and route surveys. Beginners should not book over-dimension loads without an experienced dispatcher.

The tiedown rule — and the “4 chains” myth

Securement is cargo-specific — not “4 chains per load”

FMCSA general rule for articles NOT blocked by a headboard or other cargo: minimum number of tiedowns by length.



Rule: 2 tiedowns for the first 10 ft + 1 for every extra 10 ft (or part). Total working load limit ≥ ½ the cargo weight. Commodity rules (coils, lumber, pipe, machinery...) add more. The driver secures; the dispatcher confirms the gear exists.

FMCSA's general rule, in plain words

For an article that is **not blocked** from moving forward (by a headboard or other cargo): **1 tiedown** if it is 5 ft or shorter and 1,100 lb or lighter; **2 tiedowns** if longer than 5 ft (up to 10 ft) or heavier than 1,100 lb; above 10 ft, **2 for the first 10 ft plus 1 for every extra 10 ft or part of it**. The combined working load limit of all tiedowns must be **at least half the cargo's weight**.



Worked example

A 32-ft bundle of steel beams, not blocked: first 10 ft → 2, remaining 22 ft → 3 more (10 + 10 + 2). Minimum **5 tiedowns**, and commodity rules may require more.



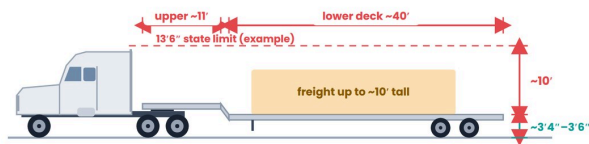
Correction: “Minimum 4 chains per flatbed load”

No such universal rule exists. The count depends on **length, weight, whether the cargo is blocked, the working load limits, and commodity-specific rules** for logs, lumber, metal coils, paper rolls, concrete pipe, vehicles, heavy machinery and boulders.

You don't need to calculate securement for the driver. You need to know enough to ask: “What does this load need, and does our truck have it?”

The step deck (drop deck)

53' STEP DECK



53-foot step deck (drop deck)

Flatbed with a short upper deck and a long lower deck. The lower deck lets taller freight travel without going over height.

Upper deck	≈ 11–13' long, about 5' high
Lower deck	≈ 40–42' long, about 3'4"–3'6" high
Freight height	≈ 10' on the lower deck (if state limit is 13'6")
Width	8'6" (102")
Cargo weight	Often ≈ 43,000–48,000 lb
Typical freight	Tall machinery, tractors, equipment, tall crates
Not the same as	Double drop / RGN / lowboy (heavy haul)

Before booking, the dispatcher asks

- Exact freight height, length, width and weight?
- Loaded height = deck height + freight height
- Ramps needed for rolling equipment?
- Any state/route with a lower height limit?

DAT equipment types: **SD** **FD**

Typical figures — they vary by manufacturer and year. * No federal height limit; states set it (usually 13'6"–14'). Confirm the actual truck with the carrier.

How to spot one

Look for the **step**: a short upper deck over the tractor, then a drop to a long, lower deck. In the photo the lower deck is loaded with long steel pieces. A **flatbed** is one flat deck at about 5 feet.

DAT has **SD** (Step Deck) and **FD** (Flatbed or Step Deck). FD postings accept either, so height may not be critical.



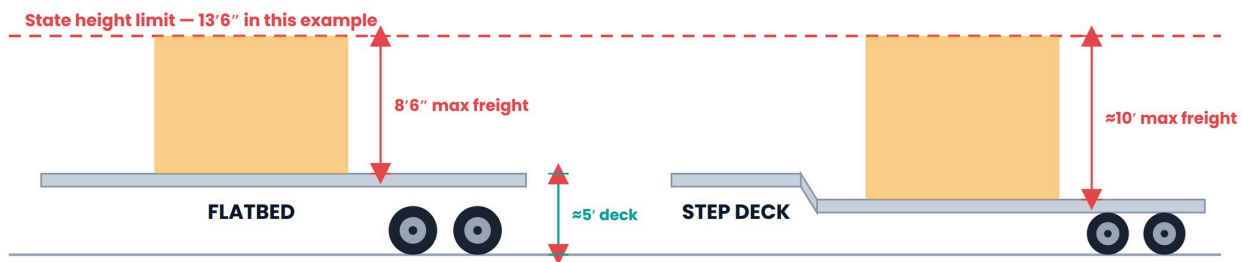
Correction: "Step deck = no permit needed for medium machinery"

A step deck **reduces loaded height**, which may keep a load legal. But permit need depends on the **total dimensions and the states on the route**. It never guarantees permit-free movement.

Flatbed or step deck? Do the height math

Why a step deck: a lower deck = taller freight under the same limit

Loaded height = deck height + freight height. A step deck helps — but never guarantees a load is permit-free.



Worked example: a 9'6" machine

On a flatbed: 5'0" deck + 9'6" = **14'6"**, over a 13'6" or 14' limit **X**. On a step deck: 3'6" + 9'6" = **13'0"**, under 13'6" **✓**, if every state and bridge on the route allows it.

Use real deck heights from the carrier, not typical ones.



Gooseneck trailer used in hotshot work

Real photo from your manual. Hotshot (pickup + gooseneck) is a growing niche. DAT lists Flatbed Hotshot (FH) and Van Hotshot (VH).

Other open-deck equipment (awareness only)

Type	What it is
Conestoga (CN)	Flatbed or step deck with a rolling tarp system, like a covered wagon
Double drop	Very low well between two raised decks, for tall freight
RGN / lowboy	Heavy haul; the front detaches so machinery can drive on
Hotshot	Pickup truck pulling a gooseneck flatbed, for smaller urgent loads



Know your limits

Heavy-haul and over-dimension freight (RGN, lowboy, permits, escorts) is a specialist job. Learn to **recognise** it and pass it on until you are trained.

Power only: the tractor without a trailer



Power only

Your carrier sends only a tractor. It hooks to a trailer supplied by the shipper, broker or another company.

What moves	Tractor only — trailer belongs to someone else
Typical use	Pre-loaded drop trailers, trailer relocation, rental trailers
Pickup style	Drop-and-hook: take a loaded trailer, leave an empty
Insurance	Carrier may need trailer interchange coverage
Paperwork	Trailer number, seal number, interchange / inspection
Watch for	Trailer condition, tires, lights, return of empties

Before booking, the dispatcher asks

- Whose trailer, what type, where is it?
- Loaded or empty? Seal number?
- Does the carrier have trailer interchange insurance?
- Where does the trailer go after delivery?

DAT equipment types: **PO**

Typical workflow — requirements vary by broker and trailer owner.



How a power-only load works

1. Your driver goes to a yard or shipper and **hooks a trailer** that is already loaded (or empty).
2. Checks the trailer: tyres, lights, brakes, doors, **seal number**. Photos before leaving.
3. Hauls it to the receiver or another yard and **drops** it.
4. Sometimes takes an **empty trailer back**. Ask whether that is paid.



Why beginners see many power-only loads

Some shippers and brokers keep their own trailer pools and just need tractors. Power only has **no trailer to own or maintain**, and it moves fast with drop-and-hook. DAT has also added power-only specialty variants in 2025.



The insurance question

When your carrier pulls someone else's trailer, the trailer owner usually wants **trailer interchange** (or similar) insurance. Confirm your carrier has it before booking.

Sprinter and cargo vans: small, fast, specific



Sprinter / cargo van

Large cargo van for urgent, small and high-value freight. Fast, flexible, usually no CDL needed.

Cargo floor	≈ 14' long (170" high-roof models)
Inside height	≈ 6'3" (75")
Between wheel wells	≈ 54–56"
Cargo weight	About 3,000–4,500 lb, depending on model
Pallets	About 2–4, limited by weight and door size
CDL	Usually not required (unless hazmat placards)
Typical freight	Expedited parts, medical, small urgent shipments

Before booking, the dispatcher asks

- Weight and pallet size — really fits?
- Liftgate or dock? Driver loads alone?
- Hazmat? Temperature-controlled? Team?
- Pickup time — expedite loads move fast

DAT equipment types: [Sprinter Van](#) [... Hazmat](#) [... Temp-Controlled](#) [... Team](#)

Figures for a high-roof 170" cargo van; other models differ.

Where cargo vans fit

Urgent, small and high-value freight: auto parts to stop a production line, medical supplies, electronics, time-critical partials. Loads are often **posted and gone within minutes**, pick up the same day, and run long distances non-stop (sometimes with a team).



Smaller vans (like this mid-size van) carry even less. Always get the real inside dimensions and payload.



Correction: "Sprinter codes are VH, MV, VL" / "MV = Mini Van"

On DAT today, **MV = Moving Van** and **VH = Van Hotshot**. DAT has explicit **Sprinter Van**, **Sprinter Van Hazmat**, **Sprinter Van Temp-Controlled** and **Sprinter Van Team** types.



Search like DAT recommends

Run a **specific** search (Sprinter Van types) and a **broad** Van Standard / Van Specialized search with **Partial** selected.

Hazmat, tankers, teams and over-dimension



- CLASS 1-Explosives
- CLASS 2-Gases
- CLASS 3-Flammable liquids
- CLASS 4-Flammable solids
- CLASS 5-Oxidizers
- CLASS 6-Toxic materials
- CLASS 7-Radioactive materials
- CLASS 8-Corrosive materials
- CLASS 9-Miscellaneous dangerous goods
- DANGEROUS- Indicates a mixed load of hazardous materials

Hazardous materials (hazmat)

Placarded hazmat needs a driver with a **hazmat endorsement** on the CDL and a carrier set up for hazmat (registration, insurance, procedures). DAT types include VZ, RZ, FZ and SB-Hazmat. **Never book hazmat unless your carrier and driver are qualified.**



Tankers

Liquids and gases in bulk. Specialised equipment, and a tanker endorsement is required for the driver. Recognise it; beginners rarely dispatch tankers.



Over-dimension

Wide, tall, long or heavy loads that need permits, sometimes escorts and specific routes. DAT: FO, LO.

Team service

Two drivers take turns driving, so the truck can move almost non-stop within hours-of-service rules. Used for urgent, long loads. DAT marks it with **VM, RM, FM** or "Team" variants. Only book team loads if your carrier really has a team on that truck.

Pallet exchange

The shipper or receiver swaps empty pallets for loaded ones. It can cost the carrier time or money if the pallets aren't returned. Ask how it works (VP, RP).



One-line rule for special equipment

If a load has an extra letter (Z, M, P, O, T...) or the word "Hazmat / Team / Over-dimension," **stop and confirm the carrier and driver are qualified and equipped** before you call.

Match the freight to the equipment

Match the freight to the equipment

Freight description → equipment → DAT equipment type. This is the first check on every load you ever consider.

 Retail pallets, 20 pallets, 38,000 lb, dock to dock	→		Dry van 53' V
 Furniture to a home, no dock, 4 pallets	→		Box truck + liftgate + pallet jack SB
 Frozen chicken, -10°F, continuous	→		Reefer R
 Steel beams 40' long, 44,000 lb	→		Flatbed + chains / straps F · FC
 Excavator 10' tall, 38,000 lb	→		Step deck (check height + ramps) SD
 Urgent car parts, 2 pallets, 1,800 lb	→		Sprinter / cargo van Sprinter
 Pre-loaded trailer waiting at a DC	→		Power only PO
 Rolls of paper that must load from the side	→		Curtainside van / flatbed VC

1 Read the freight

Commodity, weight, pallets or dimensions, temperature, hazmat.

2 Read the stops

Dock or no dock, residential, length limits, loading method (forklift, crane, side, rear).

3 Match your truck

Type, size, confirmed max weight, gear (tarps, chains, liftgate), driver qualifications.



If any of the three steps is unclear, ask before you negotiate

A rate conversation about a load your truck can't legally or physically do is wasted time, and a booked load you have to cancel damages your carrier's reputation with that broker.

Equipment types on the load board

Current DAT equipment types worth knowing

From DAT One's equipment list (checked Sep 2026). DAT lists 60+ types — learn these, then use the selector.

VANS

V	Van
VA	Van Air-Ride
VM	Van w/ Team
VC	Van w/ Curtains
VG	Van Lift Gate
VZ	Van Hazmat
VP	Van w/ Pallet Exchange
VV	Van Vented

REEFERS

R	Reefer
RA	Reefer Air-Ride
RM	Reefer w/ Team
RZ	Reefer Hazmat
RP	Reefer w/ Pallet Exchange

OPEN DECK

F	Flatbed
FT	Flatbed w/ Tarps
FC	Flatbed w/ Chains
FD	Flatbed or Step Deck
SD	Step Deck
FO	Flatbed Over Dimension
CN	Conestoga

OTHER

PO	Power Only
SB	Straight Box Truck
—	Sprinter Van (+ Hazmat / Temp / Team)
VH	Van Hotshot
MV	Moving Van

Don't memorise from old code sheets: "MV = mini van" (it is Moving Van) · "VH / MV / VL = Sprinter" · "VM = over 700 miles" · "curtain inside = V" · "VG = every box truck".

Habit: small equipment — run a specific search (Sprinter / Straight Box types) AND a broad "Partial" van search, as DAT recommends. Platforms change: re-check before Module 6.

How to learn codes the right way

- Memorise the **core**: V, VA, VM, VC, R, RA, F, FD, SD, PO, SB.
- Know that the **second letter adds a feature** (A air-ride, M team, Z hazmat, P pallet exchange, T tarps, C chains/curtains...).
- For everything else, **use the platform's equipment selector** and read the load details.



Why old code sheets are dangerous

Platforms add and rename types. DAT added explicit Sprinter, box-truck and power-only specialty types and flatbed handling attributes in 2025. A code sheet from an old PDF can make you search for the wrong truck or miss loads.

Four mistakes this module prevents

Four equipment mistakes that cost real money

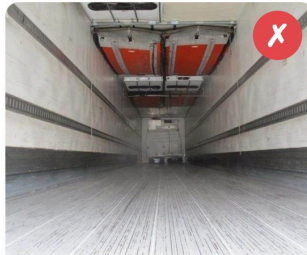
Each one means a rejected load, a TONU dispute, a claim or an angry broker. Each one is prevented by one question.



Wrong length

Receiver only fits 48'. Your 53' is turned away at the gate.

✓ Ask about length limits



Wrong temperature

Setpoint on the RC was 34°F; driver ran 0°F. Produce frozen and rejected.

✓ Read setpoint + mode back



No liftgate

Residential delivery, no dock, no liftgate. Freight cannot come off.

✓ Ask dock / liftgate / jack



Missing gear

Load needs 8' drop tarps + chains. Truck has straps only.

✓ Match gear to requirements

Mistake	What happens	The question that prevents it
Wrong length	Turned away at the gate; redelivery, maybe a TONU dispute	"Any length restriction at pickup or delivery?"
Wrong temperature	Product damaged, load rejected, claim against the carrier	"Setpoint and mode?" + read it back + check the RC
No liftgate	Freight can't be unloaded; failed delivery	"Dock or no dock? Liftgate and pallet jack needed?"
Missing gear	Driver refused at the shipper or rejected by the receiver	"What tarps, chains, straps or load bars are required?"
Overweight	Illegal truck, fines, load must be split	"Exact weight?" vs the truck's confirmed max

The questions to ask, by equipment

Equipment	Always ask	Extra checks
Every load	Commodity · weight · pallet count / pieces · dimensions if unusual · pickup and delivery type (dock / live / drop)	Special requirements in the posting notes
Dry van	Palletised or floor-loaded · stackable · length limits	Air-ride, team, load bars, straps
Box truck	Dock or no dock · residential · liftgate · pallet jack	Inside dimensions, door type, driver licence
Reefer	Setpoint · continuous or cycle · pre-cool	Product type, temp recording, clean trailer
Flatbed	L × W × H · tarps (size, drop) · chains / straps	Loading method, edge protection, over-dimension?
Step deck	Exact freight height · deck height · ramps	States on the route and their height limits
Power only	Whose trailer · loaded / empty · where it goes after	Interchange insurance, seal, trailer inspection
Sprinter	Weight · pallet size · door / liftgate	Hazmat, temp, team, same-day pickup



Say it out loud — drill 4: the equipment questions

Practise this with a broker who has just said “It’s a dry van load, Joliet to Atlanta.”

“Great. What’s the commodity and the total weight? How many pallets — are they stackable? Is it dock to dock, live load and live unload? Any length restriction, load bars or air-ride needed?”

Then do the same for a reefer (“What’s the setpoint — continuous or cycle? Pre-cool required?”) and a flatbed (“Dimensions? Tarps — what size drop? Chains or straps?”).

PRACTICE

Twelve postings: which equipment, and what do you ask?

For each posting write: **(a)** the equipment type, **(b)** the DAT type if you know it, **(c)** the one question you must ask before calling it a match. Answers at the back.

1 Des Moines, IA → Denver, CO · 22 plts · 40,500 lb
· Frozen poultry · -5°F · cycle ok

2 Charlotte, NC → Raleigh, NC · 6 plts · 7,800 lb ·
Store fixtures · Delivery: retail store, no dock

3 Birmingham, AL → Houston, TX · Steel coils ·
44,000 lb · Coil racks + chains required

4 Dallas, TX → Memphis, TN · 26 plts · 30,000 lb ·
Paper towels · Dock to dock · 53' only

5 Laredo, TX → Chicago, IL · Pre-loaded trailer at
yard · Drop & hook both ends

6 Detroit, MI → Toledo, OH · 2 plts · 1,500 lb · Auto
parts · Pick up within 1 hour

7 Peoria, IL → Omaha, NE · Wheel loader · 9'8" tall
· 34,000 lb · Ramps needed

8 Atlanta, GA → Orlando, FL · 18 plts · 38,000 lb ·
Canned food · Receiver: 48' trailers max

9 Salinas, CA → Phoenix, AZ · Lettuce · 34°F
continuous · Pre-cool required

10 Portland, OR → Boise, ID · Lumber · 46,000 lb · 8'
drop tarps required

11 Newark, NJ → Philadelphia, PA · 10 plts · 9,500 lb
· Residential delivery · Liftgate

12 Columbus, OH → Nashville, TN · 24 plts · 42,000
lb · Class 3 flammable liquids, placarded



Target

At least **10 of 12** equipment choices right, and a sensible question for each. Postings are written in the short style you will see on real load boards.

Visual quiz: name it, then ask your first question

Visual quiz: name it, then ask your first question

Eight real photos you have NOT seen in the lessons. Answers at the back.



For each photo write the equipment type and the **first question** you would ask a broker offering a load for that truck. Answers at the back.

1

2

3

4

5

6

7

8

Module 2 check

Answer on paper, then use the key. Aim for 10 of 12.

1. What is the difference between a tractor-trailer and a straight truck, and why does it matter for drop-and-hook loads?
2. The federal gross limit is 80,000 lb. Why can't a 53' dry van usually carry 80,000 – 0 lb of cargo? Give the example math.
3. How many 48" × 40" pallets fit in a 53' van straight? Turned? In a 26' box truck?
4. True or false: "The federal height limit is 13'6"." Explain.
5. A 26' box truck has a GVWR of 25,999 lb. Does the driver need a CDL to haul general freight? What if the GVWR were 33,000 lb?
6. Name four things you ask before booking a box truck to a residential address.
7. Explain continuous vs cycle mode, and who decides which is used.
8. What does FDA's rule say about pre-cooling? What was wrong with the old manual rule?
9. A 25-ft article on a flatbed, not blocked: minimum number of tiedowns under FMCSA's general rule?
10. A machine is 9'10" tall. Flatbed deck 5'0", step-deck lower deck 3'4". What is the loaded height on each, and which could be legal where the limit is 13'6"?
11. What do MV, VH, VM and VC mean on DAT today?
12. What must be true before you book a placarded hazmat load?

Answers — Module 2 check

- 1** A tractor-trailer is two units: the tractor can drop one trailer and hook another. A straight truck's box is on the same frame and can't be dropped. Drop-and-hook and power-only work need a tractor.
- 2** 80,000 lb is the gross for the whole vehicle. Example: 80,000 - tractor/fuel/driver 19,000 - trailer 15,000 = 46,000 lb on paper, and axle limits and real loading usually make it less. Ask the carrier for the truck's max.
- 3** 26 straight, up to 30 turned (if the inside width and loading allow); about 12 in a 26' box truck.
- 4** **False.** There is no federal vehicle height limit. States set limits, mostly 13'6"-14', and routes can be lower.
- 5** 25,999 lb GVWR, no hazmat → usually **no CDL**. 33,000 lb GVWR → **Class B CDL**. Length doesn't decide it.
- 6** Any four: dock or no dock; residential?; liftgate required?; pallet jack?; forklift on site?; heaviest pallet weight vs liftgate capacity; inside dimensions.
- 7** Continuous: the unit runs all the time. Cycle/start-stop: it switches off at the setpoint and restarts as needed. The shipper decides; it's usually on the RC or BOL.
- 8** The compartment must be adequately prepared, "including precooling, if necessary." The old "always 2 hours" rule is not in the regulation. Follow the shipper's instructions.
- 9** First 10 ft → 2; remaining 15 ft → 2 more (10 + 5) = **4 tiedowns** minimum, plus any commodity-specific rules.
- 10** Flatbed: 5'0" + 9'10" = **14'10" X**. Step deck: 3'4" + 9'10" = **13'2" ✓** under 13'6", if every state and bridge on the route allows it.
- 11** MV = Moving Van · VH = Van Hotshot · VM = Van w/ Team · VC = Van w/ Curtains.
- 12** The driver has a hazmat endorsement and the carrier is set up for hazmat (registration, insurance, procedures), and the truck can be placarded correctly. Otherwise, don't book.

Answers — visual quiz

#	Equipment	A good first question	#	Equipment	A good first question
1	Reefer box truck	"Setpoint and mode?"	5	Curtainside van	"Side-load required?"
2	Gooseneck flatbed (hotshot)	"Weight and dimensions?"	6	Reefer trailer (unit)	"Setpoint, continuous or cycle?"
3	Flatbed trailer	"Tarps and chains needed?"	7	Box truck	"Dock or liftgate?"
4	Small cargo van	"Real weight and size?"	8	Tanker	"Hazmat / tanker endorsed?" (specialist)

Answers — twelve postings

#	Equipment (DAT)	The question you must ask
1	53' reefer (R)	Exact setpoint and mode (cycle OK?) — and can this reefer carry 40,500 lb?
2	Box truck + liftgate (SB)	Liftgate and pallet jack needed? Heaviest fixture weight?
3	Flatbed with coil racks + chains (F · FC)	Does our truck have coil racks and enough chains? Is it legal at 44,000 lb?
4	53' dry van (V)	Floor-loaded or palletised? (26 pallets = the full floor; weight is fine.)
5	Power only (PO)	Trailer type and condition, seal, and does the carrier have trailer interchange insurance?
6	Sprinter / cargo van	Can the driver pick up within the hour? Pallet size fits the doors?
7	Step deck with ramps (SD)	Exact deck height: 3'6" + 9'8" = 13'2". States on the route? Ramps on the truck?
8	48' dry van (V)	Do we have a 48' trailer? A 53' will be refused.
9	Reefer (R), continuous	Pre-cool to what temperature? Temperature record needed?
10	Flatbed with 8' drop tarps (F · FT)	Does the truck carry lumber tarps with 8' drops? Legal at 46,000 lb?
11	Box truck with liftgate (SB)	Liftgate capacity vs pallet weight; pallet jack; residential access for the truck?
12	Hazmat van (VZ)	Is the driver hazmat-endorsed and the carrier hazmat-qualified? If not, don't book.



Notice the pattern

Almost every answer includes a **capability check** (gear, size, temperature, licence) **and** a **weight check** against the specific truck. That pattern is the habit this module builds.

Glossary

Air-ride — Suspension using air bags for a smoother ride (VA, RA).

Blocking / bracing — Wood or other material built around cargo to stop movement.

Box truck — Straight truck with an enclosed cargo box.

Bridge formula — Federal formula limiting weight on axle groups by their spacing.

Continuous mode — Reefer unit runs all the time.

Cube out — Trailer floor or space is full before reaching the weight limit.

Curtainside — Van with sliding side curtains for side loading (VC).

Cycle / start-stop — Reefer unit switches off at setpoint, restarts when needed.

Deck height — Height of a flatbed or step-deck deck above the ground.

Drop (tarp) — How far a tarp hangs down the side: 4', 6', 8'.

Drop-and-hook — Leaving one trailer and taking another instead of waiting to load.

Dunnage — Material used to fill gaps or support cargo (wood, airbags).

E-track — Metal rails on van walls for straps and load bars.

Fifth wheel — Coupling plate on the tractor that holds the trailer's kingpin.

Floor-loaded — Freight loaded loose on the floor, not on pallets.

GVWR — Gross vehicle weight rating: maximum loaded weight of a vehicle.

Kingpin — Pin under the trailer's nose that locks into the fifth wheel.

Landing gear — Legs that support a dropped trailer.

Liftgate — Powered platform that lowers freight to the ground.

Load bar — Bar wedged across a van to hold freight in place.

Over-dimension — Freight exceeding legal width, height, length or weight.

Pallet jack — Hand tool for moving pallets.

Payload — Cargo weight a truck can legally carry.

Power only — Tractor pulling a trailer it doesn't own (PO).

Pre-cool — Cooling a reefer trailer before loading, when required.

Reefer — Refrigerated, temperature-controlled trailer or truck.

Setpoint — The temperature the reefer unit is set to hold.

Step deck — Flatbed with an upper and a lower deck (SD).

Straight truck — Truck with cab and cargo body on one frame.

Tandem — Pair of axles. Trailer tandems often slide.

Tare — Empty weight of a vehicle or trailer.

Tiedown — Strap, chain or similar used to secure cargo.

Weigh out — Reaching the weight limit before the space is full.

WLL — Working load limit of a tiedown.

Sources

Publisher	Document	Where
FHWA	Vehicle Size and Weight FAQs — 80,000 lb gross, 20,000 single, 34,000 tandem, bridge formula	ops.fhwa.dot.gov/freight/sw/faqs
FHWA	Federal Size Regulations for CMVs — 102" width, 48' semitrailer minimum, no federal height limit	ops.fhwa.dot.gov/freight/publications/size_regs_final_rpt
FMCSA	Commercial Driver's License — Class A, B, C definitions	fmcsa.dot.gov/registration/commercial-drivers-license/drivers
FMCSA	Cargo Securement Rules — tiedown count, working load limit, commodity rules	fmcsa.dot.gov/regulations/cargo-securement/cargo-securement-rules
FDA	FSMA Sanitary Transportation rule — key changes (precooling "if necessary", temperature on request)	fda.gov/food/food-safety-modernization-act-fsma
DAT	DAT One Help Center — Equipment Types (60+ types incl. V, VA, VM, VC, VG, R, F, FD, SD, PO, SB, MV, VH)	one.support.dat.com
DAT	Find loads for Sprinter vans, cargo vans and box trucks (specific + broad Partial search)	one.support.dat.com
DAT	What's New — 2025 specialty equipment types and flatbed handling attributes	one.support.dat.com/whats-new
Industry	Typical trailer dimensions: Hale Trailer (dry van), O Trucking (step deck, box truck sizes), Hansatic (reefer), The Vansmith (Sprinter 170)	haletrailer.com · otrucking.com · hansatic.com · thevansmith.com
User-supplied	Dry Van & Box Truck manual; Truck Types Module 2 manual; OMNITECH manual — photos and legacy claims audited	your files



How numbers were handled

Legal limits (weight, width, length, CDL classes, securement, food transport) come from FHWA, FMCSA and FDA. Equipment dimensions and payloads are **typical industry figures** from manufacturer and industry guides. They vary by truck, so the course always says "confirm with the carrier." DAT equipment types were checked on the DAT One Help Center in September 2026.

Corrections log

Every claim from your three training manuals that this module changed, and what is taught instead.

Old manual claim	Taught instead
Dry van max payload 45,000 / flatbed 48,000 lb	Examples only — payload = legal limits - tare; ask the carrier
Mountain areas: payload 30–40k	Rejected — no general rule
Curtainside filtered under V	DAT: Van w/ Curtains (VC)
VA mandatory for fragile freight	Air-ride when the load requires it
VM = loads over 700 miles	VM = van with team
14–22 ft non-CDL / 24–28 ft CDL	CDL by GVWR (26,001 lb) and use (hazmat, passengers)
VG = box-truck liftgate code	VG = Van Lift Gate; box trucks use Straight Box Truck types
Pre-cool 2 hours before loading	Pre-cool if necessary / as instructed; verify setpoint
Minimum 4 chains per flatbed load	Cargo-specific tiedown rules + commodity rules
13'6" legal height; step deck = no permit	No federal height limit; state/route rules; permits depend on dimensions
Sprinter = VH / MV / VL; MV = mini van	DAT Sprinter Van types; MV = Moving Van; VH = Van Hotshot
Fixed \$/mile ranges per equipment	Removed — rates taught with live data in Module 7